svgBA100 FINANCIAL INTELLIGENCE PLATFORM

Comprehensive Analyst Operations & System Reference Guide

1. Executive Overview & Purpose

The **svgBA100 Financial Intelligence Platform** is an institutional-grade financial analysis, screening, valuation, and capital allocation intelligence system tailored for the Nifty 100 constituent universe. Spanning Sprints 1 through 6 (Days 1–45), the platform integrates automated ETL pipelines, authoritative SQLite data stores, deterministic ratio computation engines, natural language text extraction, machine-learning archetypal clustering, an interactive multi-page Streamlit analytical dashboard, and high-performance RESTful FastAPI services.

This guide serves as the definitive reference manual for financial analysts, portfolio managers, and quantitative engineers operating the system locally. It details architecture, screening workflows, report generation, API endpoints, troubleshooting, and documented data exceptions.

Core Platform Capabilities

- **Master Universe:** Exactly 92 verified Indian enterprise corporations with multi-year audited histories.
- **Analytics & Ratio Engine:** Deterministic computation of 55+ KPIs (ROE, ROCE, OPM, NPM, D/E, ICR, FCF, 3/5/10yr CAGRs).
- **Screener & Scoring:** Sector-relative P10/P90 normalized Composite Quality Score (0–100) and 6 predefined strategy presets.
- **Peer Cohort Benchmarking:** Percentile rankings across 11 industry cohorts and 92 automated radar charts.
- **Cash Flow Intelligence:** 8-pattern capital allocation taxonomy, distress early-warning flags, and deleveraging detectors.
- **Automated PDF Reporting:** 2-page company tear sheets, 10 broad-sector reports, and a 17-page portfolio summary.
- **Machine Learning:** 5-archetype K-Means clustering, within-sector outlier detection (|Z| > 3), and elbow analysis.
- **RESTful API & Dashboard:** 21 FastAPI endpoints under /api/v1 and an 8-screen Streamlit application.

Target Universe	Nifty 100 Constituents (92 Authoritative Master Companies)
Primary Data Store	SQLite (db/schema.sql, nifty100.db with Foreign Key Constraints ON)
Local API Server	FastAPI / Uvicorn (http://127.0.0.1:8000, OpenAPI docs at /docs)
Analytical Dashboard	Streamlit Multi-page App (http://localhost:8501)
Authoritative Sprints	Sprint 1 through Sprint 6 Complete (179 Regression Tests Passing)

2. System Architecture & Relational Data Model

The platform follows a layered modular architecture designed for local execution, deterministic auditability, and zero cloud dependency. The relational core is stored in **nifty100.db** with foreign key enforcement enabled (`PRAGMA foreign_keys = ON`).

Database Schema Overview

Table Name	Primary Key / FK	Record Count	Functional Description
companies	id (TEXT PK)	92	Master table containing company names, logos, face value, book value.
sectors	id (PK), company_id (FK)	92	10 authoritative broad sectors, sub-sectors, and index weights.
profitandloss	id (PK), company_id (FK)	1,146	Historical P&L: sales, expenses, operating profit, net profit, EPS.
balancesheet	id (PK), company_id (FK)	1,143	Historical BS: equity, reserves, borrowings, total assets/liabilities.
cashflow	id (PK), company_id (FK)	1,091	Historical CF: operating (CFO), investing (CFI), financing (CFF).
financial_ratios	id (PK), company_id (FK)	1,160	Calculated ratios: ROE, ROCE, OPM, D/E, ICR, CAGRs, quality score.
peer_groups	id (PK), company_id (FK)	59	11 industry peer cohorts with designated benchmark tickers.
peer_percentiles	id (PK), company_id (FK)	560	Peer cohort percentiles for 10 financial metrics.
market_cap	id (PK), company_id (FK)	552	Market cap, enterprise value, PE, PB, EV/EBITDA, dividend yield.
documents	id (PK), company_id (FK)	1,457	Annual report PDF URLs and regulatory filing links.

Authoritative Broad Sectors vs Industry Peer Cohorts

A critical architectural distinction verified during Sprint 5 and preserved in Sprint 6 is that the database maintains **two complementary categorization dimensions**:

1. **10 Authoritative Broad Sectors:** Populated in the `sectors` table (Communication Services, Consumer Discretionary, Consumer Staples, Energy, Financials, Healthcare, Industrials, Information Technology, Materials, Real Estate). Every one of the 92 companies belongs to exactly one broad sector.
2. **11 Industry Peer Cohorts:** Populated in `peer_groups` (Automobiles, Consumer Finance, FMCG, IT Services, Life Insurance, Oil & Gas, Pharmaceuticals, Power & Utilities, Private Banks, Public Sector Banks, Steel). Used specifically for intra-industry percentile rankings.

Analysts must not conflate these dimensions: `/api/v1/sectors` exposes the 10 broad sectors, while `/api/v1/peers` exposes the 11 peer cohorts.

3. Dashboard Navigation & Screen Walkthrough (Part 1)

The interactive Streamlit dashboard is launched via `streamlit run src/dashboard/app.py --server.port 8501`. The application provides eight dedicated analytical screens accessible via the left sidebar.

Screen 01: Home & Executive Overview

The Home screen displays high-level portfolio KPIs: total universe market capitalization, average ROE, median Debt-to-Equity, and top-ranked companies by Composite Quality Score. It features a distribution chart of the 10 broad sectors and direct quick-links to detailed company analyses.

Screen 02: Company Profile & Financial Statements

Allows single-company drill-down via ticker selection. Displays:

- Master corporate profile: sector, sub-sector, face value, book value, and website.
- Key Financial Ratios: ROE %, ROCE %, Net Profit Margin %, OPM %, Debt-to-Equity, and ICR.
- Full historical tables for Profit & Loss, Balance Sheet, and Cash Flows with annual/TTM toggles.
- Automated Strengths and Weaknesses generated from the rule-based NLP engine.
- Direct one-click download of the 2-page publication-grade PDF Tear Sheet.

Screen 03: Multi-Metric Financial Screener

The core screening module allowing interactive filtering across 15 metrics. Analysts can activate one of the 6 predefined presets or customize minimum/maximum thresholds for ROE, ROCE, D/E, FCF, Revenue CAGR, and Valuation multiples. Results update instantaneously with sorting on any metric and CSV export functionality.

Screen 04: Peer Comparison & Benchmarking

Enables head-to-head benchmarking within any of the 11 industry peer cohorts. Displays side-by-side metric tables, percentile rankings (0–100th percentile), benchmark company deltas, and embedded 10-axis radar charts generated during Sprint 3.

4. Dashboard Navigation & Screen Walkthrough (Part 2)

Screen 05: Historical Trends & Growth Analysis

Focuses on multi-year longitudinal trends. Provides interactive dual-axis line charts comparing revenue growth vs operating profit, net margin progression, and compound annual growth rates (3-year, 5-year, and 10-year CAGRs). Highlights inflection points and growth decelerations.

Screen 06: Broad Sector Intelligence

Summarizes sector-level fundamental health across all 10 authoritative broad sectors. Features median ROE, ROCE, leverage, and market capitalization concentration charts. Clicking any sector reveals its constituent companies and sector-relative rankings.

Screen 07: Capital Allocation & Cash Flow Health

Visualizes the 8-pattern cash flow taxonomy established in Sprint 5. Identifies whether companies are Shareholder Returners, Reinvestors, Cash Accumulators, or Distress Signals. Highlights early warning distress alerts (negative CFO funded by CFF borrowings) and active deleveraging trends.

Screen 08: Institutional Reports & Document Hub

Central repository for viewing and downloading institutional PDF deliverables: individual company tear sheets, 10 broad-sector reports, the 17-page portfolio summary document, and direct hyperlinks to official annual report filings stored in the database.

Operational Summary of Dashboard Screens

Screen	Module Path	Key Analytical Function	Primary Data Source
01 Home	src/dashboard/pages/01_home.py	Portfolio snapshot & market overview	companies, sectors, financial_ratios
02 Profile	src/dashboard/pages/02_profile.py	Company financials, pros/cons, PDF export	All financial tables, prosandcons
03 Screener	src/dashboard/pages/03_screener.py	15-metric filtering & composite scoring	src/screener/engine.py
04 Peers	src/dashboard/pages/04_peers.py	11 peer cohorts, radar charts & ranks	peer_groups, peer_percentiles
05 Trends	src/dashboard/pages/05_trends.py	CAGR time series & margin trends	financial_ratios, analysis
06 Sectors	src/dashboard/pages/06_sectors.py	10 broad sectors median analysis	sectors, financial_ratios
07 Capital	src/dashboard/pages/07_capital.py	8-pattern cash flow & distress alerts	output/cashflow_intelligence.xlsx
08 Reports	src/dashboard/pages/08_reports.py	PDF report repository & regulatory links	reports/, documents table

5. Financial Screener & Composite Quality Scoring

The platform includes an institutional-grade screener operating on `src/screener/engine.py`. The engine evaluates all 92 companies against custom threshold constraints and calculates a sector-relative Composite Quality Score.

Composite Quality Score Methodology (0–100 Scale)

To prevent cross-sector bias (e.g. IT services naturally having higher ROE than capital-heavy utilities), every component metric is normalized sector-relative using 10th and 90th percentile winsorisation:

```
$$Score_{i} = \frac{\text{clip}(X_i, P_{10}, P_{90}) - P_{10}}{P_{90} - P_{10}} \times 100$$
```

The four foundational pillars and their exact scoring weights are:

1. **Profitability (35%):** Return on Equity (15%), Return on Capital Employed (10%), Net Profit Margin (10%).
2. **Cash Quality (30%):** 5-Year FCF CAGR (15%), CFO/Operating Profit Quality Ratio (10%), Positive FCF Bonus (5%).
3. **Growth (20%):** 5-Year Compounded Revenue CAGR (10%), 5-Year PAT CAGR (10%).
4. **Leverage & Solvency (15%):** Debt-to-Equity (10%, inverted), Interest Coverage Ratio (5%). Financial sector companies are exempted from standard industrial D/E penalties.

Standard Screener Presets

Preset Key	Preset Strategy Name	Target Investment Criteria
quality_compounders	Quality Compounder	ROE > 15%, D/E <= 1.0 (excl. Financials), FCF > 0, Rev CAGR 5yr > 10%.
value_picks	Value Pick	P/E <= 20.0, P/B <= 4.5, D/E <= 2.0, Dividend Yield >= 1.0%.
growth_accelerators	Growth Accelerator	PAT CAGR 5yr > 20%, Revenue CAGR 5yr > 15%, D/E <= 2.0.
dividend_champions	Dividend Champion	Dividend Yield >= 2.0%, Payout Ratio < 80%, Free Cash Flow > 0.
debt_free_blue_chips	Debt-Free Blue Chip	D/E <= 0.05, ROE > 12.0%, Annual Sales > 5,000 Cr.
turnaround_watch	Turnaround Watch	Revenue CAGR 3yr > 10%, Positive Latest FCF, Declining D/E YoY.

6. Institutional PDF Reports & Automated Generation

Sprint 5 established automated, publication-quality PDF reporting using ReportLab. All generated documents adhere strictly to A4 page geometry with zero blank pages, text clipping, or overlapping canvas elements.

1. Individual Company Tear Sheets (reports/tearsheets/*.pdf)

Each eligible company possesses an authoritative **exactly two-page** PDF tear sheet (file size 47KB–118KB):

- **Page 1:** Navy corporate header bar, 6 KPI dashboard tiles (ROE, ROCE, NPM, D/E, Rev CAGR, FCF), 10-year Revenue & Net Profit bar chart, and 10-year ROE/ROCE dual-axis trend line chart.
- **Page 2:** Balance sheet composition stacked bar chart (Equity vs Borrowings vs Other Liabilities), latest cash-flow waterfall chart (CFO, CFI, CFF, Net), rule-based Strengths & Weaknesses bullet points, and Capital Allocation Pattern badge.
- **Eligible Coverage:** 91 companies. **JIOFIN** is documented as skipped due to having only 2 annual observations (<3 required).

2. Broad-Sector Intelligence Reports (reports/sector/*.pdf)

A dedicated one-page PDF report for each of the **10 authoritative broad sectors**. Each report presents sector median ROE, ROCE, OPM, D/E, 5yr Rev CAGR, total market cap, and a constituent fundamentals table.

3. Executive Portfolio Summary (reports/portfolio/portfolio_summary.pdf)

A comprehensive **17-page A4 document** indexing all 92 master companies alphabetically by ticker. Each company card provides: broad sector, fiscal year basis, market cap, portfolio weight, 6 core KPIs, capital allocation pattern, cash-flow quality label, and 5-year Revenue and PAT CAGR trend direction indicators (Up / Down / Stable).

7. Statistical Clustering & Company Archetypes

Implemented in `src/analytics/clustering.py`, the statistical clustering pipeline categorizes all 92 companies into five distinct financial archetypes using unsupervised machine learning.

Clustering Methodology & Preprocessing

- **Feature Selection (5 KPIs):** Return on Equity (%), Debt-to-Equity, 5-Year Revenue CAGR (%), 5-Year FCF CAGR (%), Operating Profit Margin (%).
- **Missing Data Imputation:** Missing values are imputed using the company's broad-sector median. If an entire sector lacks data (e.g. FCF CAGR in Communication Services), the portfolio-wide median is used as a deterministic fallback.
- **Feature Scaling:** Standardized using `StandardScaler` (zero mean, unit variance).
- **K-Means Algorithm:** Initialized with `n_clusters=5`, `random_state=42`, and Euclidean distance to centroid computed for every company.

Five Provisional Cluster Archetypes

Cluster ID	Archetype Designation (Provisional)	Count	Profile Summary & Representative Members
Cluster 0	High-Margin Cash Generators	14	Very high operating margin (median 85.1%), negligible debt (median 0.05). Members: ADANIP
Cluster 1	Core Quality Compounders	60	Steady revenue growth (11.4%), solid profitability (ROE 17.0%), modest debt (0.17). Members
Cluster 2	High Capital Turnover & Scale	2	PSU defense manufacturing champions with extreme ROE and capital turnover. Members: BE
Cluster 3	Hyper FCF Outlier	1	Pharmaceutical giant exhibiting hyper-expansion in 5-year free cash flow CAGR (228.8%). Me
Cluster 4	High-Leverage Financials & Utilities	15	High financial leverage (D/E mean 7.2) and robust revenue CAGR (19.0%). Members: HDFCB

Statistical Artifacts Generated

- `output/cluster_labels.csv`: 92 rows mapping `company_id`, `cluster_id`, `cluster_name`, and `distance_from_centroid`.
- `output/outlier_report.csv`: 9 flagged observations where within-sector absolute Z-score > 3.0 (e.g. INDIGO, PNB, BAJAJHLDNG).
- `output/portfolio_stats.csv`: P10, P25, P50, P75, P90, mean, and standard deviation across primary portfolio KPIs.
- `reports/elbow_plot.png` & `reports/correlation_heatmap.png`: Visual statistical diagnostic figures.

8. RESTful API Architecture & Endpoint Reference

Implemented in `src/api/main.py`, the FastAPI server provides 21 endpoints under `/api/v1`. It features automated request timing/logging middleware, CORS headers for frontend integration, and interactive OpenAPI docs at `/docs`.

Method	Endpoint Path	Description	Status Codes
GET	/api/v1/health	API health & database connectivity check	200, 503
GET	/api/v1/health/counts	Table row counts across all SQLite tables	200, 503
GET	/api/v1/companies	List universe companies with sector and index weight	200
GET	/api/v1/companies/{ticker}	Detailed company profile, ratios, pros/cons	200, 404
GET	/api/v1/companies/{ticker}/pnl	Historical profit & loss statements	200, 404
GET	/api/v1/companies/{ticker}/balancesheet	Historical balance sheet statements	200, 404
GET	/api/v1/companies/{ticker}/cashflow	Historical cash flow statements	200, 404
GET	/api/v1/companies/{ticker}/ratios	Complete historical financial ratios	200, 404
GET	/api/v1/companies/{ticker}/tearsheet	Download official 2-page PDF tear sheet	200, 404
GET	/api/v1/screener/presets	List predefined screener presets	200, 500
GET	/api/v1/screener	Run financial screener with presets or custom filters	200, 400
GET	/api/v1/sectors	List 10 authoritative broad sectors & medians	200
GET	/api/v1/sectors/{sector_name}	Broad sector detail & constituent companies	200, 404
GET	/api/v1/peers	List 11 industry peer cohorts & benchmarks	200
GET	/api/v1/peers/{group_name}	Peer group members & percentile rankings	200, 404
GET	/api/v1/companies/{ticker}/peers	Company peer group benchmarking & percentiles	200, 404
GET	/api/v1/valuation/summary	Universe valuation summary (PE, PB, EV/EBITDA)	200
GET	/api/v1/companies/{ticker}/valuation	Company valuation history & sector multiples	200, 404
GET	/api/v1/portfolio/stats	Portfolio KPI distribution stats (P10..P90)	200, 503
GET	/api/v1/portfolio/clusters	K-Means archetypes & centroid distances	200, 400, 503
GET	/api/v1/companies/{ticker}/documents	Official annual report filing URLs	200, 404

9. Practical API Code Examples (Python & cURL)

Starting the API Server Locally

```
uvicorn src.api.main:app --host 127.0.0.1 --port 8000 --reload
```

Example 1: Running the Quality Compounder Screener via Python

```
import requests url = "http://127.0.0.1:8000/api/v1/screener" params = {"preset": "quality_compounders",
"limit": 5} response = requests.get(url, params=params) data = response.json() print(f"Matched
{data['total_matched']} companies:") for co in data['results']: print(f"- {co['company_id']}
({co['company_name']}): Score {co['composite_quality_score']}")
```

Example 2: Downloading a Company PDF Tear Sheet via cURL

```
curl -X GET http://127.0.0.1:8000/api/v1/companies/TCS/tearsheet --output TCS_tearsheet.pdf
```

Example 3: Querying Portfolio Clustering Archetypes

```
import requests url = "http://127.0.0.1:8000/api/v1/portfolio/clusters" res = requests.get(url,
params={"cluster_id": 1}).json() print(f"Cluster 1 count: {len(res['assignments'])}") for row in
res['assignments'][:5]: print(f" {row['company_id']}: dist={row['distance_from_centroid']}")
```

10. Troubleshooting & Documented Data Exceptions

Institutional integrity requires complete transparency regarding source data anomalies, edge cases, and known exceptions. The following findings were documented across Sprints 1–5 and verified in Sprint 6:

Documented Platform Exceptions

Issue / Exception	Root Cause & Source Evidence	Platform Resolution & System Behavior
JIOFIN Skipped Tearsheet	JIOFIN has only 2 annual P&L observations in source data (demerged entity).	Logged to `output/skipped_tearsheets.csv`; API returns 404 with descriptive e
SBIN & SIEMENS No-Con Trigger	SBIN has no raw balance sheet rows; SIEMENS triggered extrapolation outlier thresholds.	Documented in `output/qa_unresolved.csv`; both companies retain va
10 Sectors vs 11 Peer Cohorts	Sectors table has 10 broad economic sectors; peer analysis uses 11 industry cohorts.	Preserved both distinctions: /api/v1/sectors returns 10, /api/v1/peers
INFY CAGR Flag for Review	10.2 percentage point divergence between text-parsed and ledger-calculated CAGR.	Logged to `output/cagr_cross_validation.csv` as FLAG_FOR_REVIEW withou
WIPRO Parser Failure	WIPRO text appears in source filings but WIPRO is not in the 92 master companies.	Documented in `output/parse_failures.csv` as an unmapped ticker.
Cash Accumulator Zero Count	No company in the latest annual period simultaneously had negative CFO, +CFI, & +CFF.	Tax-haven CFO + CFI + CFF patterns; distribution CSV correctly reflects 0 count.

Common Operational Troubleshooting

- **Port Collisions:** Ensure Streamlit runs on port 8501 and FastAPI on port 8000. Verify with `netstat -ano | findstr :8000`.
- **Database Locks:** SQLite operates in WAL or standard rollback mode. Do not perform external write transactions while running batch ETL.
- **Pytest HTML Reports:** Generated via `pytest --html=reports/pytest_report.html --self-contained-html`.

11. Production Operations Runbook & Audit Sign-Off

This section outlines standard procedures for maintaining, testing, and auditing the platform in local production environments.

Essential Execution Commands

```
# 1. Run Complete Automated Regression Test Suite python -m pytest --html=reports/pytest_report.html
--self-contained-html
# 2. Execute Statistical Clustering & Analysis Pipeline python -m src.analytics.clustering
# 3. Launch Local RESTful API Server uvicorn src.api.main:app --host 127.0.0.1 --port 8000
# 4. Launch Interactive Analytical Dashboard streamlit run src/dashboard/app.py --server.port 8501
# 5. Run Performance & Integration Benchmarks python scripts/benchmark_perf.py
```

Verification Gate Status & Team Approvals

All 20 Sprint 6 acceptance criteria (AC-01 through AC-20) have been technically audited and verified. This guide, the underlying software artifacts, and the test reports provide factual evidence of system behavior. Final operational deployment and cluster archetype designations remain **Provisional** pending formal team-lead review and sign-off.

Role	Designation	Verification Status	Date
Implementation Lead	Senior Python Agentic Engineer	COMPLETED (Technically Verified)	2026-10-23
Quantitative Reviewer	Financial Data Lead	PROVISIONAL (Pending Review)	Outstanding
Product Owner / Team Lead	svgBA100 Project Director	PROVISIONAL (Pending Sign-off)	Outstanding